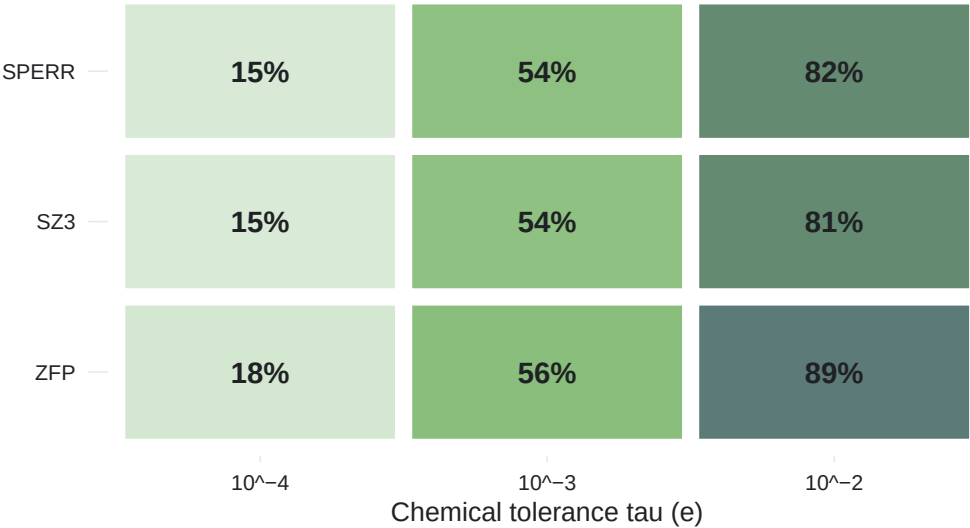


Figure 3 | Chemical certification is a stability-aware decision problem

Compression fidelity is scientifically interpretable only after the downstream observable is identifiable at the requested tolerance.

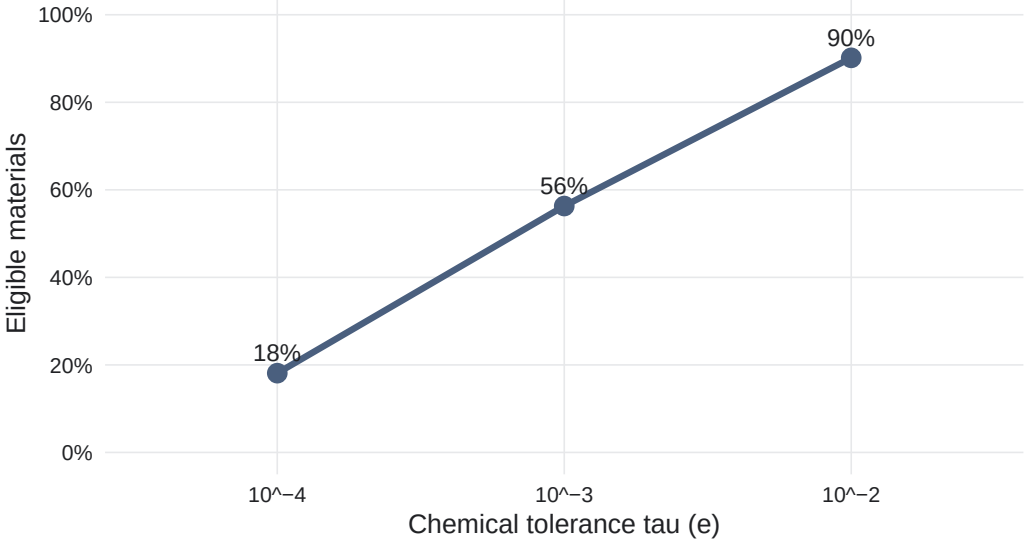
A | Certification depends on tolerance and codec

Protocol A.1 eligible + re-derived Bader error below tau



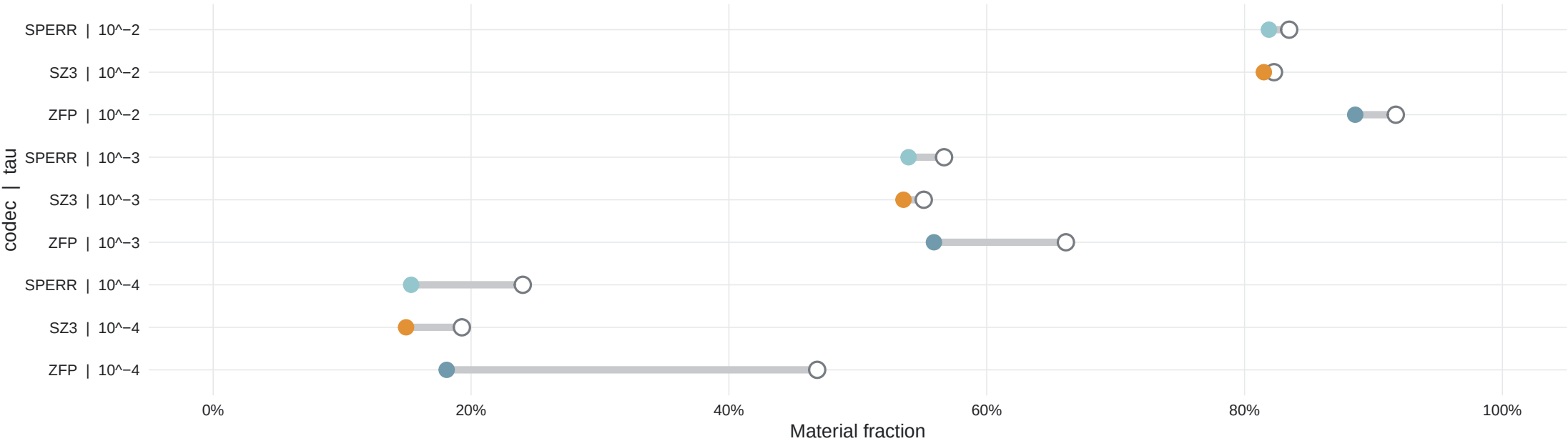
B | The A.1 stability floor sets the identifiability ceiling

Eligibility is defined before codec success is counted



C | Ignoring eligibility systematically overstates scientific success

Filled: Protocol A.1 certified; open: Bader criterion counted without eligibility



Development corpus. Protocol A.1 uses fixed-seed uniform perturbations at the float32 L-infinity amplitude. Certification requires both eligibility and re-derived Bader error below the chemical tolerance.